



SMRFORGE

VERIFIABLE SMR COMPUTATIONAL ENGINEERING

smrforge.io · Open-core · Community open source · Pro in beta

SMR DESIGN · ANALYSIS · OPTIMIZATION

Reactor physics your reviewers can reproduce — not just believe.

A full screening reactor-physics engine for Small Modular Reactors where every result ships as a fingerprinted, independently reproducible evidence bundle — and is labeled, honestly, for how far to trust it.

```
git clone → pip install → python demo/walkthrough.py → a screening k-eff
                                                    you verify in a
                                                    minute
```

SMRFORGE.EVIDENCE_ENVELOPE.V1 ◆ SEALED

analysis	smr-core-screening
k_eff	1.29841
fidelity	screening
sha-256	a3f1c8...9e0b
replay	bit-exact @ epoch

```
$ smrforge verify ./bundle
✓ sealed · schema ok · unedited · exit 0
```

THE DIFFERENTIATOR — BUILT FOR THE PERSON WHO DOESN'T TRUST YOU

01 · FINGERPRINT

Evidence envelopes

Every output is a schema-registered bundle sealed by a SHA-256 manifest of its inputs and result — any edit breaks the seal, and it is **ed25519**-signable (a demo identity today, production key on the roadmap).

02 · REPLAY

Bit-exact reproduction

A pinned `SOURCE_DATE_EPOCH` and the `reproduce_bundle()` Python API regenerate the same bytes from the same inputs.

03 · VERIFY

exit 0 = it holds

A standalone, offline check against the open **Evidence Envelope** schema — a third party confirms your record's integrity (unedited and schema-valid) or refutes it, in seconds, no SMRForge license needed (re-running the number end-to-end is the separate REPLAY step above). Digests also enter an append-only Merkle log: inclusion and consistency proofs it was never rewritten.

ONE REPRODUCIBLE MODEL, A FULL SCREENING REACTOR-PHYSICS STACK — ACROSS LWR-SMR, HTGR (PRISMATIC & PEBBLE), MSR & MICROREACTOR ARCHETYPES

NEUTRONICS	Full-core nodal & NEM · CE Monte Carlo & S _n transport	KINETICS	Point-kinetics · reactivity coefficients	FUEL CYCLE	Burnup & depletion · Xe/Sm poisoning
SAFETY	Rod-worth & shutdown-margin · pin-power	UNCERTAINTY	Data-uncertainty · sensitivity	NUCLEAR DATA	In-tree ENDF/B · self-shielding, TSL
GEOMETRY	Cartesian cores · R-Z / hex on engagement	EVIDENCE	Sealed bundles · offline verify · transparency log	INTEROP	OpenMC round-trip · Serpent/MCNP converters on engagement

HONEST BY DEFAULT · OPEN-CORE

WHAT IT GIVES YOU

- ✓ A fidelity label + plausibility band on every result
- ✓ Machine-readable caveats that travel with the output
- ✓ One-command escalation to OpenMC (free) for a reference-grade k-eff

WHAT IT WON'T PRETEND

- Screening k-eff is a trend, not a validated absolute
- The open core is honestly screening-grade, not licensing sign-off
- Bundles seal under a published demo identity — a check proves integrity, not origin
- Nothing is tuned to hit a benchmark it was measured against

Community smrforge · MIT

The full screening engine — prove fit in the open.

- Full screening stack (nodal + NEM, CE-MC, S_n, burnup, kinetics) + evidence spine & reviewer-kit
- Archetype presets (LWR-SMR, HTGR, MSR, micro) → a runnable, baseline-checked model via the Python API
- One-step OpenMC escalation, free (Serpent/MCNP converters on engagement)
- Benchmarks, stable Python API; CLI ships verify and --version today

Pro smrforge_pro · FSL-1.1-MIT COMMERCIAL

The assurance / regulatory layer for submissions.

- The signed assess deliverable — a real reference solve + a tech-spec shutdown-margin gate in one bundle, with a PASS/FAIL verdict against the safety floor
- Requirements-to-evidence traceability · SRP-aligned report artifacts
- Confirmatory Console — re-run a submission, diff every claim, ship a sealed verification report; subsidized for regulators & national labs
- Advanced geometry (R-Z, hex) & full-core CMFD available on engagement

```
# Screening k-eff + live verify, from a checkout: $ python demo/walkthrough.py # → computes k, verifies it with a standalone checker, then catches a signed lie
```

Open standard & verifier →

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